

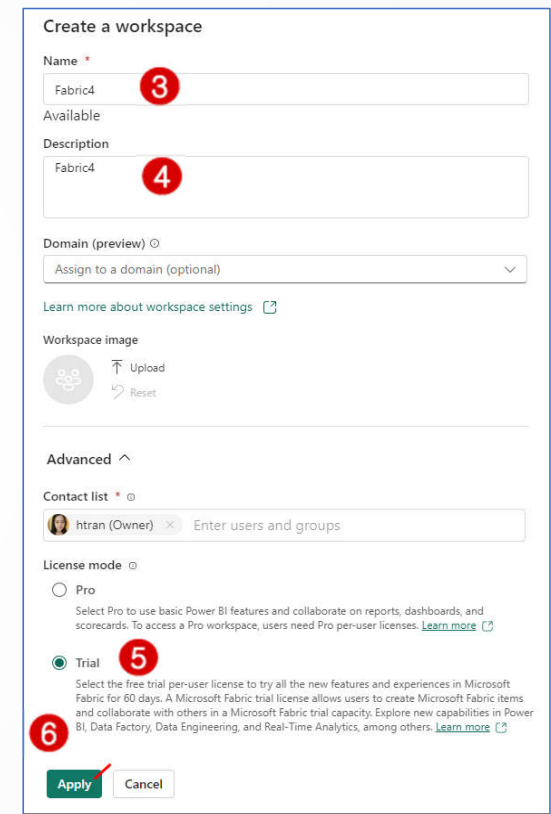
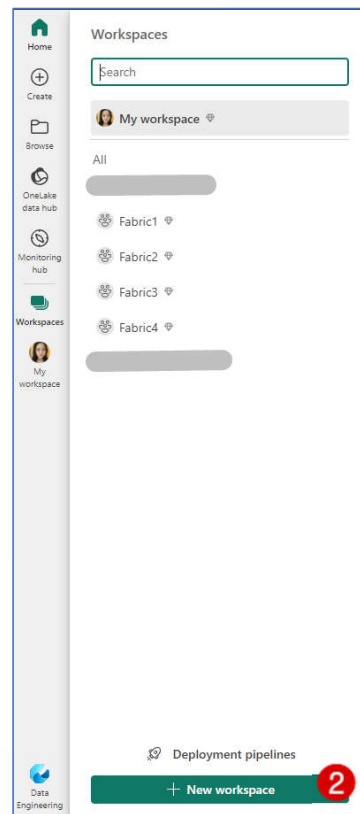
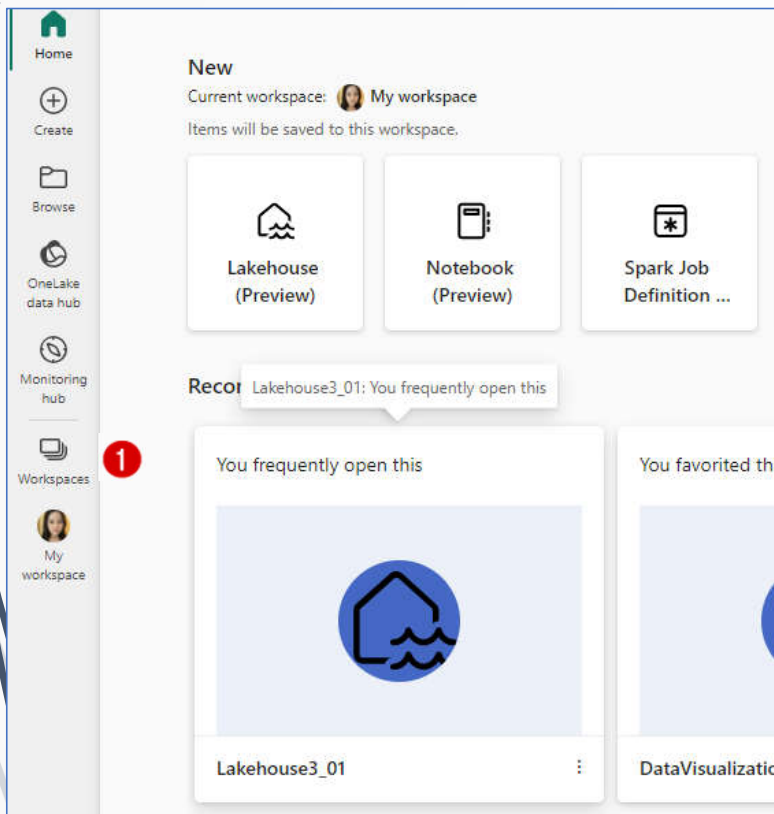
MS Fabric (Preview)



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1.Create Workspaces

➤ [Synapse Data Engineering \(microsoft.com\)](https://synapse.dataengineering.microsoft.com)



2. Adding Lakehouse

Lakehouse is built into the Fabric service and provides a unified location to store all organizational data where the experiences operate

The screenshot shows the Microsoft Fabric interface. On the left, the 'New' menu is open, and 'Lakehouse (Preview)' is highlighted with a red box and a red circle with the number 1. In the center, the 'New' section shows a grid of options, with 'Lakehouse (Preview)' highlighted by a red box and a red circle with the number 2. On the right, the 'New lakehouse' dialog box is open, showing the name 'lakehouse1' entered in the 'Name' field, highlighted by a red box and a red circle with the number 3. The 'Create' button is highlighted by a red box and a red circle with the number 4. Below the main interface, a diagram titled 'OneSecurity' shows the 'OneLake' architecture. It features a central 'OneLake' logo with 'Delta - Parquet Format' below it. To the left are 'Warehouse' and 'Lakehouse', both with 'Delta - Parquet Format' below them. To the right are 'Kusto DB' and 'Dataset', both with 'Delta - Parquet Format' below them.

Microsoft Fabric →

- Power BI
- Data Factory
- Synapse
- Data Engineering **1**
- Data Science
- Data Warehouse
- Real-Time Analytics

New

Current workspace: Fabric4
Items will be saved to this workspace.

- Lakehouse (Preview) 2**
- Notebook (Preview)
- Spark Job Definition ...
- Data pipeline (Preview)
- Import notebook
- Use a sample

Recommendations Fabric3: You frequently open this

- You frequently open this: Fabric3
- You frequently open this: Lakehouse3_01
- You frequently open this: Fabric1

New lakehouse

Name *

lakehouse1 **3**

Create 4 Cancel

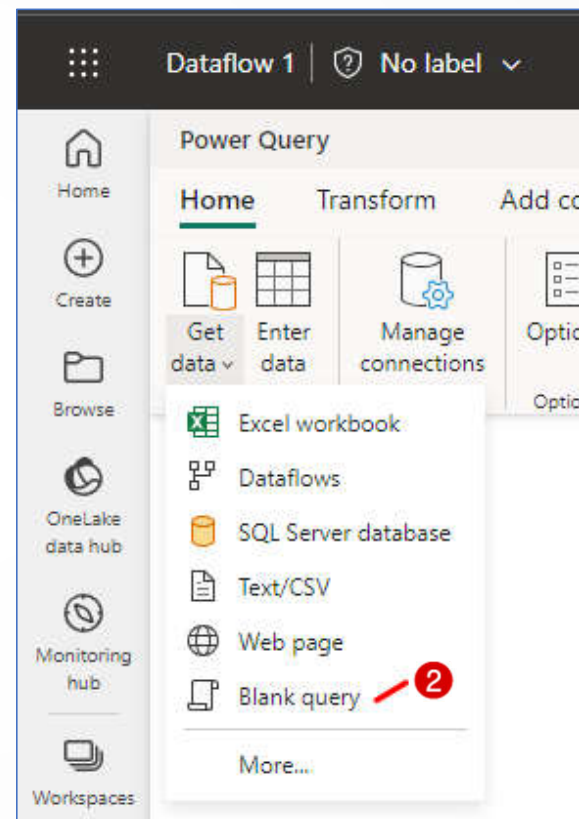
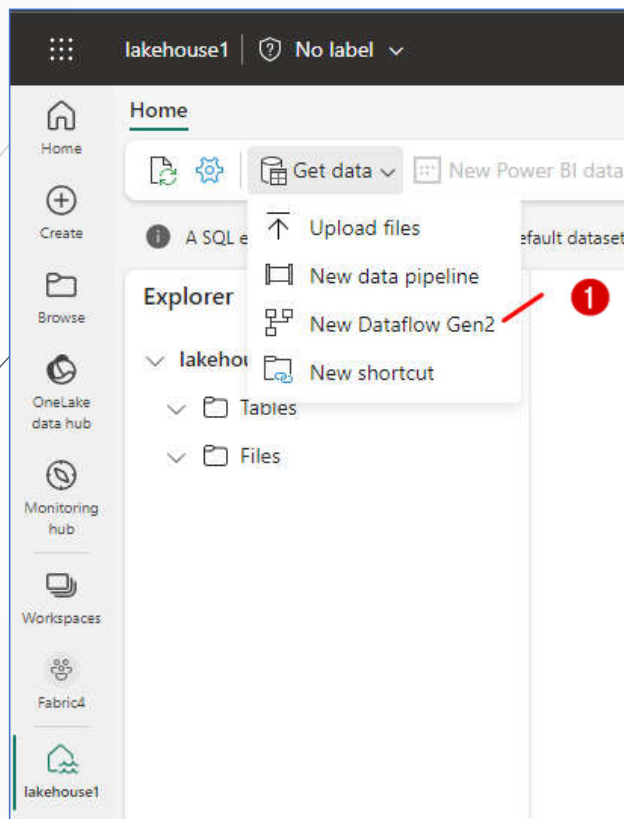
OneSecurity

Warehouse **Lakehouse** **OneLake** **Kusto DB** **Dataset**

Delta - Parquet Format **Delta - Parquet Format** **Delta - Parquet Format** **Delta - Parquet Format**

3. Dataflow Gen2

Create Date dim table by using query



3. Dataflow Gen2 > Get data

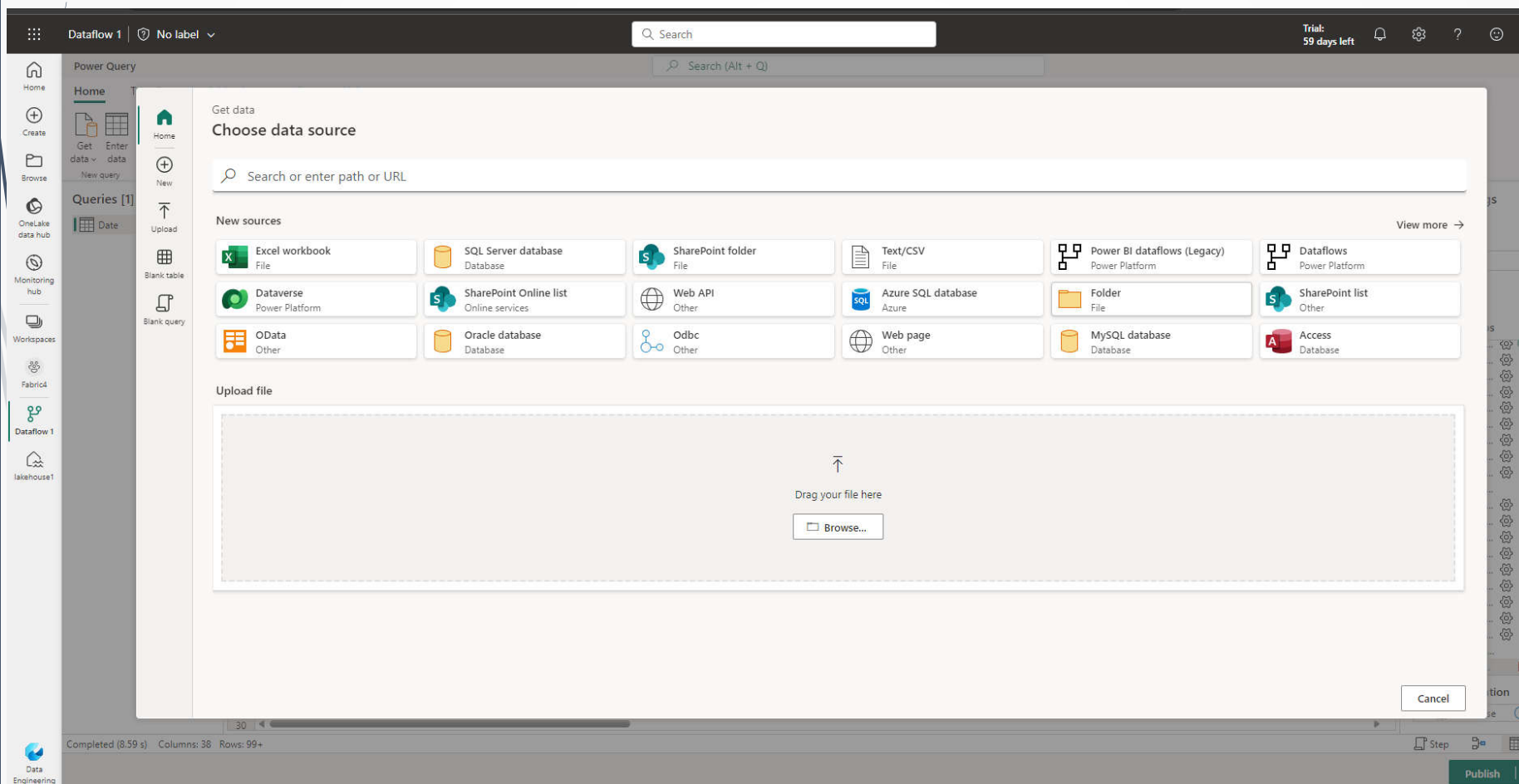
Input query and rename the table after creation

The screenshot shows the 'Get data' interface in Dataflow Gen2. The 'Connect to data source' panel on the left has a 'Blank query' button. The main 'Query' panel contains a complex MQL script. A red arrow points to the 'let' statement at the top of the script. A red circle with the number '3' is placed over the 'Next' button at the bottom right of the 'Get data' panel.

The screenshot shows the 'Transform' interface in Dataflow Gen2. The 'Queries' panel on the left shows a list of queries. A red circle with the number '4' is placed over the 'Run' button next to the first query. The 'Query settings' panel on the right shows the 'Name' and 'Query' fields.

3. Dataflow Gen2 > get data

Choose data from many sources



3. Dataflow Gen2 > get data

Upload files > Select tables to get data

Get data

Connect to data source

Excel workbook

File

[Learn more](#)

Connection settings

☐ Link to file ☒ Upload file (Preview) 1

Sales_Data 3.xlsx

Upload successful (1.7 MB)

Connection credentials

Connection

Authentication kind: Organizational account [Edit connection](#)

Dataflow 1 | No label

Search

Power Query

Home

Get data

Enter data

Queries [1]

Get data

Choose data

Display options

Excel workbook

Customer

Geography

Item

Sales 2

OrderID	ItemID	SalesDate	DeliveryDate	Customer	CityID	Qty	Price	Cost	DiscountPercent
20277	37	10/5/2020	10/6/2020	2691	26	1	233	172	12
22328	23	10/5/2020	10/7/2020	8083	61	1	60	54	9
22784	53	10/5/2020	10/9/2020	3562	19	1	333	259	5
25881	14	10/5/2020	10/9/2020	338	32	2	753	301	39
26399	52	10/5/2020	10/13/2020	4993	51	2	53	45	28
26895	51	10/5/2020	10/10/2020	2651	33	2	73	36	36
29167	33	10/5/2020	10/13/2020	1002	51	1	267	202	45
26273	32	10/5/2020	10/9/2020	4796	73	2	91	77	20
26728	46	10/2/2020	10/7/2020	5137	4	1	83	65	27
21121	53	10/2/2020	10/5/2020	386	33	2	233	172	10
21136	14	10/2/2020	10/11/2020	1162	89	1	63	52	32
21320	32	10/2/2020	10/10/2020	1679	62	2	40	36	13
22361	26	10/2/2020	10/9/2020	2954	32	2	83	66	29
23121	53	10/2/2020	10/10/2020	2023	93	2	91	77	25
23166	20	10/2/2020	10/10/2020	862	27	1	47	39	1
25941	40	10/2/2020	10/4/2020	3869	77	1	40	38	13
26619	29	10/2/2020	10/7/2020	1070	69	2	91	77	2
29171	49	10/2/2020	10/3/2020	1763	87	2	97	67	44
29189	17	10/2/2020	10/5/2020	5063	63	2	47	39	45
29191	28	10/2/2020	10/8/2020	4356	72	2	67	53	14
29236	32	10/2/2020	10/12/2020	1123	30	1	83	70	45
29969	15	10/2/2020	10/11/2020	441	63	2	267	202	12
20994	51	10/1/2020	10/9/2020	2004	83	1	333	259	12
21021	33	10/1/2020	10/6/2020	3651	45	2	63	70	16
21025	18	10/1/2020	10/8/2020	4123	96	1	40	38	5
21449	19	10/1/2020	10/5/2020	5592	35	2	65	52	36
21926	55	10/1/2020	10/1/2020	6017	86	2	200	140	9

Back

Cancel Create 3

Completed (0.58 s) Columns: 10 Rows: 99

Data Engineering

3. Dataflow Gen2 > Transform data, set data destination

Using power query like power BI to transform data (Change data types, add new columns...)

The screenshot displays the Microsoft Fabric Dataflow Gen2 interface. The main workspace shows a Power Query transformation step named 'Table.RenameColumns(#"Changed Type1", ...)'. The transformation is applied to a table with columns: Date, StartOfMonth, EndOfMonth, MonthYear, MonthYearSort, StartOfWeek, EndOfWeek, StartOfYear, EndOfYear, StartOfQtr, EndOfQtr, and QtrYear. The data is organized into a grid with 30 rows and 12 columns. A red dashed box highlights the transformation step, and a red arrow points to the 'Data destination' dropdown menu in the 'Applied steps' pane, which is set to 'Lakehouse'. The 'Query settings' pane on the right shows the 'Properties' tab with 'Name' and 'Date' fields. The 'Applied steps' pane shows a list of steps, with the current step highlighted. The 'Data destination' dropdown is labeled with a red '4'. The 'Connection credentials' pane shows the connection to 'Lakehouse (none)' with 'Authentication kind: Organizational account'. The 'Connection credentials' pane is labeled with a red '5'. The 'Data destination' dropdown is labeled with a red '6'. The 'Data destination' dropdown is labeled with a red '4'. The 'Data destination' dropdown is labeled with a red '4'.

Columns: 38 Rows: 99+

3. Dataflow Gen2 > set data destination

Set destination target to publish data

Choose destination target

New table

Existing table

Search

Display options

Lakehouse

Fabric1

Fabric2

Fabric3

Fabric4

DataflowsStagingLakehouse

lakehouse1

My workspace

Back

A new table will be created in lakehouse1

Table name*

Date

Choose destination settings

Update method

Existing data

New data

Append

Replace

Column mapping

Source	Source type	Destination	Destination type
☒ FYDayofYear	1^3 Whole number	FYDayofYear	Whole number
☒ FyMonth	1^2_3 Whole number	FyMonth	Whole number
☒ WeekofYear	1^2_3 Whole number	WeekofYear	Whole number
☒ YearWeek	1^2_3 Whole number	YearWeek	Whole number
☒ FYWeekYearStart	Date/Time	FYWeekYearStart	Date/Time
☒ FYWeek	1^2_3 Whole number	FYWeek	Whole number
☒ FYYearWeek	1^2_3 Whole number	FYYearWeek	Whole number
☒ FYQtrNo	1^2_3 Whole number	FYQtrNo	Whole number
☒ FYQtrStartDate	Date/Time	FYQtrStartDate	Date/Time
☒ FYEndofQtr	Date/Time	FYEndofQtr	Date/Time
☒ FYQtr	1^2_3 Whole number	FYQtr	Whole number
☒ QtrFY	Text	QtrFY	Text

Back

Cancel

Save settings

5

3. Dataflow Gen2 > Publish

Set destination target to publish data

The screenshot displays the Microsoft Power Query interface within the Dataflow Gen2 environment. The main window shows a data table with 30 rows and 11 columns. The columns are: OrderNo, ItemID, SalesDate, DeliveryDate, CustomerId, CityId, Qty, Price, Cost, DiscountPercent, and Gross. The data represents sales transactions over time.

The interface includes a ribbon with the following tabs: Home, Transform, Add column, View, and Help. The 'Add column' tab is currently selected, showing various options like 'Index column', 'Rank column', 'Duplicate column', 'Merge columns', 'Extract', 'Format', 'Parse', 'Cluster values', 'Statistics', 'Standard Scientific', 'Trigonometry', 'Rounding', and 'Information'.

On the left side, there is a 'Queries' list with 5 queries: Customer, Date, Geography, Item, and Sales. The 'Sales' query is selected.

On the right side, the 'Query settings' panel is visible, showing the 'Properties' tab. The 'Name' is 'Sales', the 'Entity type' is 'Custom', and the 'Applied steps' list includes 'Source', 'Navigation 1', 'Promoted h...', 'Changed co...', 'Added cust...', and 'ABC 123 Changed co...'. The 'Data destination' is set to 'Lakehouse'.

At the bottom of the interface, there is a status bar indicating 'Completed (3.71 s) Columns: 11 Rows: 99+'. A 'Publish' button is visible in the bottom right corner.

4. Using SQL endpoint in Fabric

The image shows two screenshots of the Microsoft Fabric interface, illustrating the steps to use the SQL endpoint.

Top Screenshot: The 'Home' page of the Fabric workspace. A red circle '1' highlights the 'SQL endpoint' button in the top right corner, which is labeled 'Query data using SQL'.

Bottom Screenshot: The 'Get data in your lakehouse' page. A red circle '2' highlights the 'New SQL query' button. A red circle '3' highlights the SQL query editor, which contains the following query:

```
--SQL Query 1
1 SELECT a2.Category, SUM(Gross)
2 FROM Sales a1
3 LEFT JOIN Item a2 ON a1.ItemID=a2.ItemID
4 GROUP BY a2.Category
5 ORDER BY a2.Category
```

A red circle '4' highlights the 'Run' button. Below the query editor, the 'Results' tab shows a table with 5 rows and 2 columns: 'ABC Category' and '12L_unstried2'.

ABC Category	12L_unstried2
Category 1	2302872
Category 2	3335090
Category 3	1448671
Category 4	2388220
Category 5	802955

5. Create model

The screenshot displays the Microsoft Fabric Lakehouse1 interface, specifically the 'Create model' process. The interface is divided into several sections:

- Explorer (Left Panel):** Shows the hierarchy of the lakehouse, including Warehouses, Schemas, and Tables. The 'Customer' table is highlighted in the 'dbo' schema.
- Home (Top Bar):** Contains navigation links for Home, Reporting, and Table tools. A search bar is also present.
- Diagram (Central Workspace):** A visual representation of the data model. It shows five tables: Customer, Sales, Item, Date, and Geography. The tables are connected by lines indicating relationships. Red numbered circles 2, 3, 4, and 5 highlight the connections between the tables.
- Table Details (Right Panel):** Displays the columns and data types for the selected table. The 'Sales' table is currently selected, showing columns like CityId, Cost, CustomerId, DeliveryDate, DiscountPercent, Gross, ItemId, OrderNo, Price, Qty, and SalesDate.

Red numbered circles 1 through 5 highlight key steps in the 'Create model' process:

1. Selecting the 'Model' tab in the bottom navigation bar.
2. Connecting the 'Customer' table to the 'Sales' table.
3. Connecting the 'Item' table to the 'Sales' table.
4. Connecting the 'Geography' table to the 'Sales' table.
5. Connecting the 'Date' table to the 'Sales' table.

6. Create measure for using in reports

The screenshot displays the Microsoft Fabric Lakehouse interface, specifically the 'Measure tools' tab. The interface is divided into several sections:

- Top Bar:** Shows the lakehouse name 'lakehouse1' and a search bar.
- Left Sidebar:** Contains navigation icons for Home, Create, Browse, OneLake data hub, Apps, Metrics, Monitoring hub, Deployment pipelines, Learn, Workspaces, Fabric4, and lakehouse1.
- Explorer Panel:** Lists tables and measures. The 'Tables' section includes Customer, Date, Item, and Geography. The 'Measures' section lists Gross_MTD, Budget, Discount, Profit, and Margin. The 'Gross_MTD' measure is highlighted with a red circle 3.
- Model View:** A star schema diagram showing the relationships between tables. The 'Sales' table is the central fact table, connected to 'Customer' (1 to many), 'Date' (1 to many), 'Item' (1 to many), and 'Geography' (1 to many). The 'Sales' table has columns: CityId, Cost, CustomerId, DeliveryDate, DiscountPercent, Gross, ItemId, OrderNo, Price, Qty, and SalesDate.
- Measure Tools Bar:** Includes a 'New measure' button (highlighted with a red circle 1), a data category dropdown (set to 'Uncategorized'), and a formula bar (highlighted with a red circle 2) containing the DAX formula: `Gross_MTD = CALCULATE(SUM(Sales[Gross]), DATESMTD('Date'[Date]))`.

7. Report

The image displays the Microsoft Power BI interface, split into two main sections: the workspace on the left and a report on the right.

Left Panel (lakehouse1 workspace):

- Home:** Includes buttons for 'New SQL query', 'New visual query', 'New report' (highlighted with a red circle and arrow), and 'New measure'.
- Explorer:** A sidebar showing a hierarchy of data sources:
 - Warehouses:** A single item.
 - Tables:** A list of tables including 'Customer' and 'Item'.
 - Customer:** Fields include Age, City, CustomerId, Name, and State.
 - Item:** Fields include Brand, BrandId, Category, CategoryId, ItemId, Name, SubCategory, and SubCategoryId.
 - Sales:** A list of sales-related metrics including CityId, Cost, CustomerId, DeliveryDate, and DiscountPercent.

Right Panel (Sales dashboard report):

- Visuals:** The dashboard contains four main visualizations:
 - Profit by State:** A map of North America showing profit distribution by state.
 - Profit by Category:** A treemap showing profit distribution across five categories (Category 1 to Category 5).
 - Margin by EndOfMonth:** A line chart showing the margin over time from January 2019 to July 2020.
 - Sum of Gross by Brand:** A bar chart showing the sum of gross for various brands.
- Filters:** A sidebar on the right of the report area with sections for 'Filters on this visual', 'Filters on this page', and 'Filters on all pages'. It includes a search bar and checkboxes for various fields like 'Category', 'Profit', 'City', 'State', 'Brand', and 'Sales'.
- Visualizations:** A section on the far right with a 'Build visual' button and a list of available visual types (e.g., Bar chart, Line chart, Map, Treemap).

8. Notebook

The screenshot displays the Databricks Lakehouse environment. On the left, a sidebar contains navigation icons for Home, Create, Browse, OneLake data hub, Monitoring hub, Workspaces, Fabric4, and PowerBI4. The main area is divided into three sections: 'New' (Lakehouse and Notebook previews), 'Recommended' (Lakehouse3_01), and 'Lakehouse explorer' (showing a tree view of tables and files). The central notebook editor is titled 'DataVisualization Copy' and shows a Python script for data manipulation and visualization. The script includes imports for numpy, pandas, wordcloud, seaborn, and matplotlib. It defines a data list and creates a 1D array, then transforms it into a 2D array. The output of the script is displayed below the code.

DATA MANIPULATION AND VISUALIZATION WITH PYTHON

```
1 import numpy as np
2 import pandas as pd
3 from wordcloud import WordCloud, STOPWORDS
4 import seaborn as sns
5 import matplotlib.pyplot as plt
6 import folium
```

1. Numpy Array

```
1 # Declaration data variable
2 data = [30, 22, 47, 20, 46, 47, 37, 24, 40, 12, 43, 45, 44, 48, 24, 30, 15, 28, 21, 29, 17, 24, 41, 33, 23]
3 print('Số lượng phần tử trong data list:', len(data))
4
5 #Create 1 dim array
6 arr = np.array(data)
7 print('\nMảng 1 chiều arr:', arr, '\n')
8
9 row = len(data)//5
10
11 #Transform 1 dim array to 2 dim array
12 arr.shape=(int(row),5)
13 arr
```

Số lượng phần tử trong data list: 25

9. Pipeline

Building pipeline for load resource automatically from external database

